Supplementary Appendix

This appendix has been provided by the authors to give readers additional information about their work.

Supplement to: Shitara K, Bang Y-J, Iwasa S, et al. Trastuzumab deruxtecan in previously treated HER2-positive gastric cancer. N Engl J Med 2020;382:2419-30. DOI: 10.1056/NEJMoa2004413

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List of Investigators

Country, Study Site	Principal Investigator
Japan	
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Iwate Medical University Hospital	Keisuke Koeda
Miyagi Cancer Center	Yasuko Murakawa
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National Cancer Center Hospital East	Kohei Shitara
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Tokyo Metropolitan Komagome Hospital	Yasushi Omuro
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Kitasato University Hospital, The Kitasato Institute	Kenji Ishido
Kanagawa Cancer Center	Takashi Oshima
Niigata Cancer Center Hospital	Hiroshi Yabusaki
Shizuoka Cancer Center	Hirofumi Yasui
Aichi Cancer Center Hospital	Kei Muro
Osaka University Hospital	Daisuke Sakai
Local Incorporated Administrative Agency, Osaka City Hospital Organization, Osaka City General Hospital	Kohei Akiyoshi
Kindai University Hospital	Hisato Kawakami

Kobe City Medical Center General Hospital	Hisateru Yasui
Hyogo Cancer Center	Masahiro Tsuda
Okayama University Hospital	Toshiyoshi Fujiwara
Kagawa University Hospital	Akihito Tsuji
National Hospital Organization Shikoku Cancer Center	Tomohiro Nishina
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National Hospital Organization Kyushu Cancer Center	Taito Esaki
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Toranomon Hospital	Yuji Miura
Japan Organization of Occupational Health and Safety, Kansai Rosai Hospital	Atsushi Takeno
Shizuoka General Hospital	Keisei Taku
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Toyonaka Municipal Hospital	Hiroshi Imamura
Hiroshima Prefectural Hospital	Katsunori Shinozaki
Tochigi Cancer Center	Hisayuki Matsushita
Hiroshima City Asa Citizens Hospital	Jun Hihara
Fukui Prefectural Hospital	Tamon Miyanaga
Kanazawa University Hospital	Takeshi Terashima
Japanese Red Cross Kyoto Daini Hospital	Naoki Kakihara
Ibaraki Prefectural Central Hospital	Kenji Amagai
Yokohama City University Medical Center	Chikara Kunisaki
National Hospital Organization Kure Medical Center	Hirotaka Tashiro

Korea	
Seoul National University Hospital	Do-Youn Oh
Yonsei Cancer Center, Severance Hospital, Yonsei University Health System	Hyun Cheol Chung
Seoul National University Bundang Hospital	Keun-Wook Lee
Samsung Medical Center	Jeeyun Lee
Asan Medical Center	Min-Hee Ryu
Korea University Anam Hospital	Yeul Hong Kim
Chonnam National University Hwasun Hospital	Woo Kyun Bae
Chonbuk National University Hospital	Eun Kee Song
The Catholic University of Korea, Seoul St. Mary's Hospital	In-Ho Kim
Korea University Guro Hospital	Sang Cheul Oh
Gangnam Severance Hospital, Yonsei University Health System	Jae-Yong Cho
Chungbuk National University Hospital	Yaewon Yang
Kyungpook National University Chilgok Hospital	Jong Gwang Kim
National Cancer Center	Young Iee Park
Inje University Haeundae Paik Hospital	Sung Sook Lee
Chung-Ang University Hospital	In Gyu Hwang
Dong-A University Hospital	Sung Yong Oh
Gachon University Gil Medical Center	Sun Jin Sym

Supplementary Methods

Inclusion Criteria

- Men or women aged ≥20 years
- Pathologically documented locally advanced or metastatic gastric or gastroesophageal junction adenocarcinoma
    - Centrally confirmed human epidermal growth factor receptor 2 overexpression (immunohistochemistry [IHC] 3+ or IHC2+/in situ hybridization positive) on archival tissue (or fresh sample if archived tissue is inadequate)
    - Progression on and after ≥2 prior regimens that included fluoropyrimidine, a platinum agent, and trastuzumab or approved trastuzumab biosimilar
        - Patients who have previously progressed on either irinotecan or paclitaxel (physician's choice of drug) can receive the other drug they did not progress on if deemed appropriate by the treating physician
    - Measurable disease based on Response Evaluation Criteria in Solid Tumors version 1.1
- Patients with a primary tumor accessible by endoscopy agree to submit fresh tumor samples for HER2 assessment before registration
- Eastern Cooperative Oncology Group performance status of 0 to 1
- Left ventricular ejection fraction ≥50%
- Platelet count ≥100 × 10^9/L
- Hemoglobin ≥8.0 g/dL
- Absolute neutrophil count ≥1.5 × 10^9/L
- Creatinine clearance ≥30 mL/min
- Alanine aminotransferase/aspartate aminotransferase ≤3 × upper limit of normal (ULN)
- Total bilirubin ≤1.5 × ULN
- Albumin ≥2.5 g/dL
- International normalized ratio and activated partial thromboplastin time ≤1.5 × ULN

Exclusion Criteria

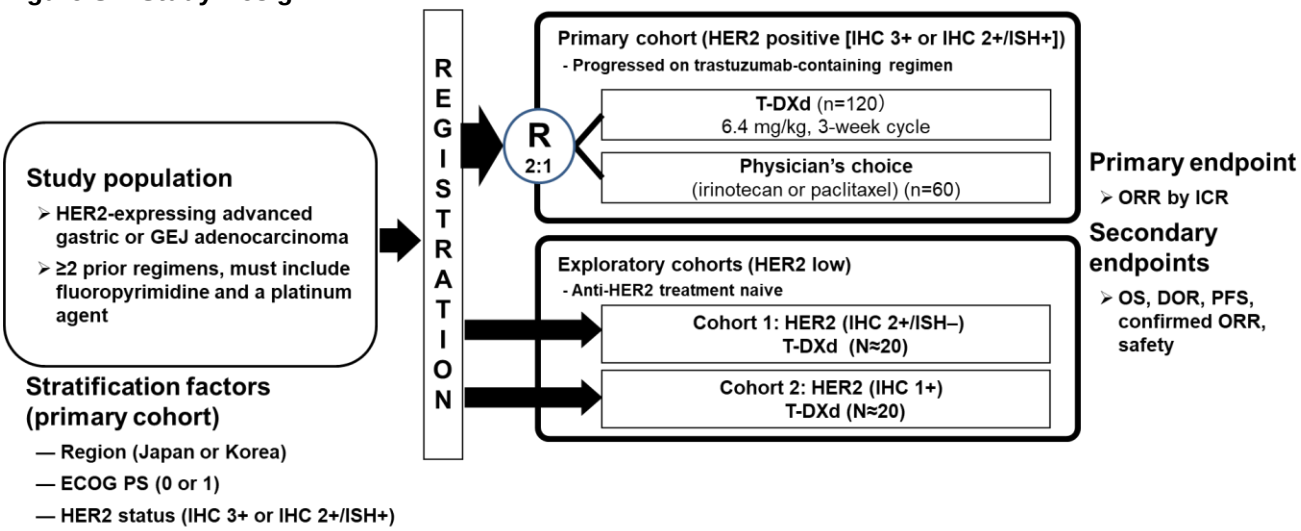
- Myocardial infarction ≤6 months before registration, symptomatic congestive heart failure, troponin levels consistent with myocardial infarction, unstable angina, or serious cardiac arrhythmia requiring treatment within 28 days of randomization/registration
- Corrected QT interval prolongation >470 ms (women) or >450 ms (men)
- History of (noninfectious) interstitial lung disease (ILD)/pneumonitis that required steroids, current ILD/pneumonitis, or suspected ILD/pneumonitis that cannot be ruled out by imaging at screening
- Pleural effusion, ascites, or pericardial effusion requiring drainage, peritoneal shunt, or cell-free and concentrated ascites reinfusion therapy
- Uncontrolled infection requiring intravenous injection of antibiotics, antivirals, or antifungals
- HIV+
- Active hepatitis B or C
- Brain metastases that are untreated, symptomatic, or require treatment to control symptoms
- Clinically significant corneal disease
- Prior treatment with an antibody-drug conjugate containing an exatecan derivative that is a topoisomerase I inhibitor
- Unresolved toxicities from prior anticancer therapy
- Diarrhea, ileus, jaundice, or intestinal paralysis
- A concomitant medical condition that could increase toxicity
- Hypersensitivity to study treatment or its inactive ingredients
- History of severe hypersensitivity to monoclonal antibodies
- Nongastric or gastroesophageal junction primary malignancies in ≤3 years except resected nonmelanoma skin cancer, curatively treated in situ disease, and other curatively treated solid tumors
- Lactating or pregnant

Safety Assessment

The safety assessment included analysis of treatment-emergent adverse events (AEs), serious AEs, AEs leading to treatment discontinuation, AEs of special interest (AESI), and AESI leading to treatment discontinuation. Additional safety assessments included analysis of physical examination findings, vital sign measurements, standard clinical laboratory parameters, elevated troponin levels, electrocardiogram parameters, echocardiogram/multigated acquisition scan findings, and ophthalmologic findings. Patients receiving trastuzumab deruxtecan were monitored for antidrug antibodies.

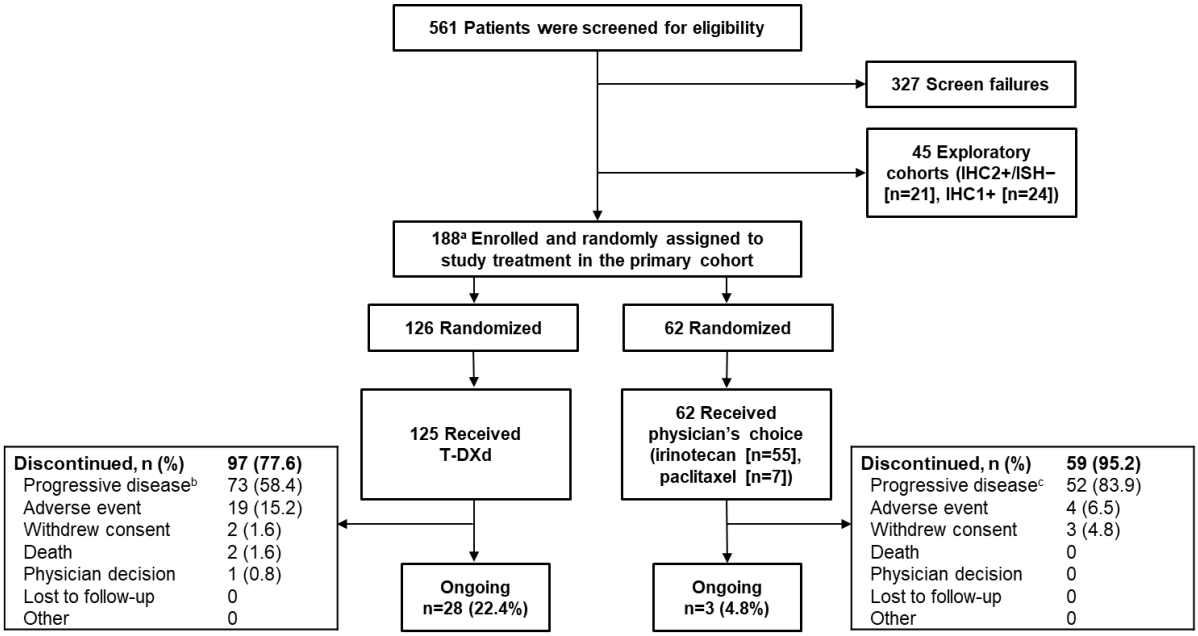
Supplementary Tables and Figures

Figure S1. Study Design.



DOR, duration of response; EGOG PS, Eastern Cooperative Oncology Group performance status; GEJ, gastroesophageal junction; HER2, human epidermal growth factor receptor 2; ICR, independent central review; IHC, immunohistochemistry; ISH, in situ hybridization; ORR, overall response rate; OS, overall survival; PFS, progression-free survival; T-DXd, trastuzumab deruxtecan.

Figure S2. Patient Disposition.



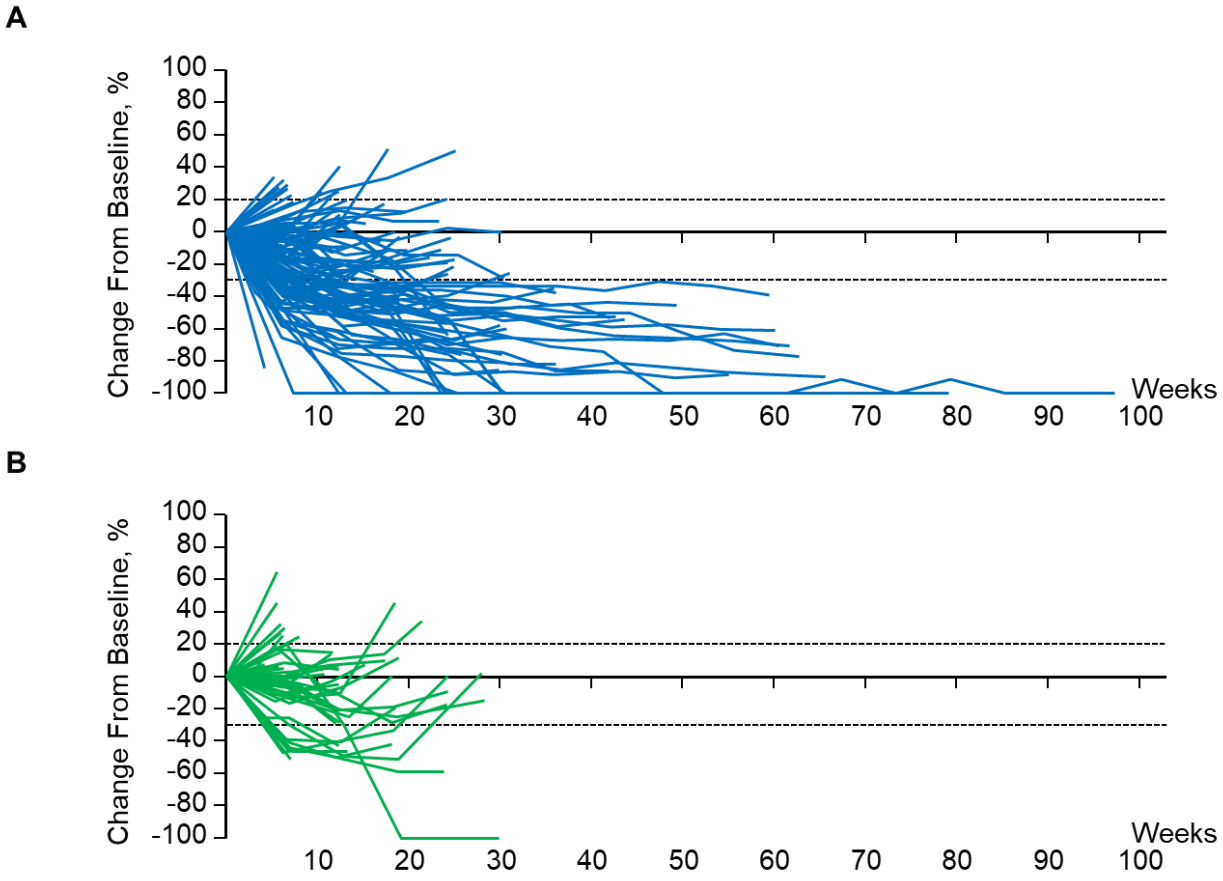
T-DXd, trastuzumab deruxtecan.

[a] One patient was not eligible because informed consent was not obtained prior to study procedures being performed. An additional 45 patients who met the eligibility criteria were enrolled and were included in exploratory cohorts.

[b] Includes patients who discontinued due to progressive disease (n=67) and clinical progression (n=6).

[c] Includes patients who discontinued due to progressive disease (n=47) and clinical progression (n=5).

Figure S3. Spider Plot (changes from baseline in tumor burden, measured as the sum of tumor diameters) in (A) T-DXd Group and (B) Physician's Choice Group.

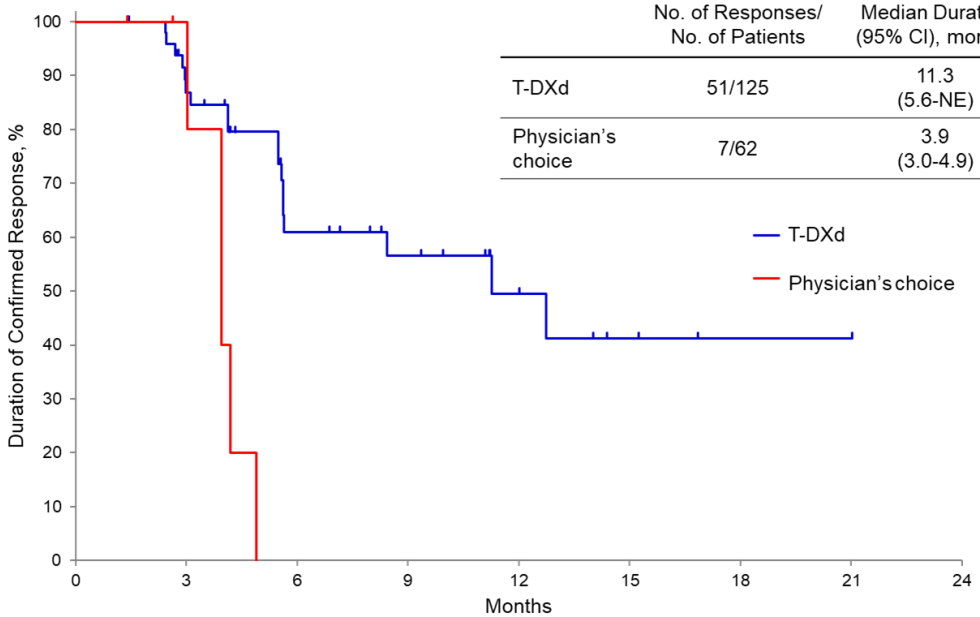


T-DXd, trastuzumab deruxtecan.

Baseline is defined as the last measurement taken before the first dose of study drug.

Dashed lines denote 20% increase or 30% decrease in tumor size.

Figure S4. Kaplan-Meier Analysis of Duration of Confirmed Response Based on ICR.



	No. of Responses/ No. of Patients	Median Duration (95% CI), months
T-DXd	51/125	11.3 (5.6-NE)
Physician's choice	7/62	3.9 (3.0-4.9)

No. at risk

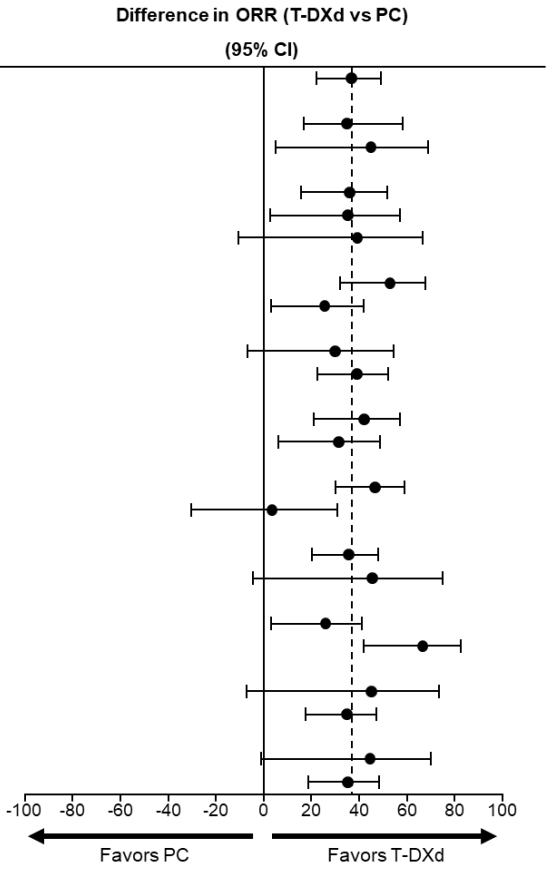
T-DXd	**51**	**37**	**19**	**13**	**7**	**3**	**1**	**1**	**0**
Physician's choice	**7**	**5**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

NE, not estimable; T-DXd, trastuzumab deruxtecan.

Duration of confirmed response among patients with confirmed complete response or partial response (irinotecan, n=6; paclitaxel, n=1; trastuzumab deruxtecan [T-DXd], n=51). In the T-DXd arm, 33 patients (64.7%) were censored; in the physician's choice arm, two patients (28.6%) were censored.

Figure S5. Subgroup Analysis of Difference in ORR by ICR With T-DXd and Physician's Choise.

Subgroup Name	T-DXd			PC		
	N	*Responders*	*%*	*N*	*Responders*	*%*
Primary cohort	119	61	51.3	56	8	14.3
Region						
Japan	95	48	50.5	45	7	15.6
Korea	24	13	54.2	11	1	9.1
Lines of prior systemic therapy						
2	60	29	48.3	33	4	12.1
3	34	18	52.9	17	3	17.6
≥4	25	14	56.0	6	1	16.7
Age						
<65 years	52	30	57.7	22	1	4.5
≥65 years	67	31	46.3	34	7	20.6
Sex						
Female	30	14	46.7	12	2	16.7
Male	89	47	52.8	44	6	13.6
ECOG PS						
0	59	31	52.5	29	3	10.3
1	60	30	50.0	27	5	18.5
HER2 status in central laboratory						
Primary cohort: IHC 3+	91	53	58.2	44	5	11.4
Primary cohort: IHC 2+/ISH+	28	8	28.6	12	3	25.0
Primary tumor location						
Gastric	104	52	50.0	49	7	14.3
GEJ	15	9	60.0	7	1	14.3
Histological subtype						
Intestinal	86	41	47.7	37	8	21.6
Diffuse	27	18	66.7	14	0	0.0
No. of metastatic sites						
<2	21	13	61.9	6	1	16.7
≥2	98	48	49.0	50	7	14.0
Previous total gastrectomy						
Yes	21	12	57.1	8	1	12.5
No	98	49	50.0	48	7	14.6

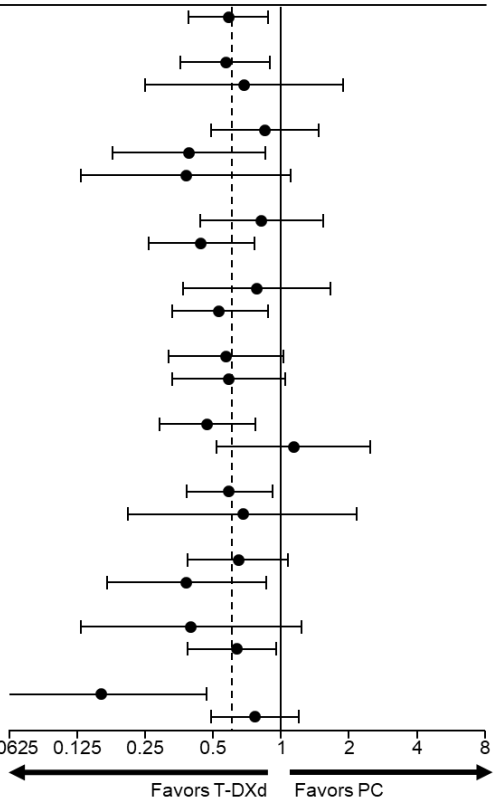


ECOG PS, Eastern Cooperative Oncology Group performance status; GEJ, gastroesophageal junction; HER2, human epidermal growth factor receptor 2; HR, hazard ratio; ICR, independent central review; IHC, immunohistochemistry; ISH, in situ hybridization; ORR, objective response rate; PC, physician's choice; T-DXd, trastuzumab deruxtecan.

Data shown for difference in ORR are for patients in the response-evaluable set (all randomized patients who received ≥1 dose of study drug and had measurable tumors based on independent central review at baseline).

Figure S6. Subgroup Analyses of OS With T-DXd and Physician's Choice.

Subgroup Name	N	No. of Events (%)	HR (95% CI)
Primary cohort	187	101 (54.0)	0.59 (0.39-0.88)
Region			
Japan	149	81 (54.4)	0.57 (0.36-0.89)
Korea	38	20 (52.6)	0.69 (0.25-1.88)
Lines of prior systemic therapy			
2	104	55 (52.9)	0.85 (0.49-1.47)
3	52	29 (55.8)	0.39 (0.18-0.85)
≥4	31	17 (54.8)	0.38 (0.13-1.11)
Age			
<65 years	82	45 (54.9)	0.82 (0.44-1.53)
≥65 years	105	56 (53.3)	0.44 (0.26-0.76)
Sex			
Female	45	30 (66.7)	0.78 (0.37-1.66)
Male	142	71 (50.0)	0.53 (0.33-0.87)
ECOG PS			
0	92	48 (52.2)	0.57 (0.32-1.02)
1	95	53 (55.8)	0.59 (0.33-1.04)
HER2 status in central laboratory			
Primary cohort: IHC 3+	143	70 (49.0)	0.47 (0.29-0.77)
Primary cohort: IHC 2+/ISH+	44	31 (70.5)	1.14 (0.52-2.50)
Primary tumor location			
Gastric	163	89 (54.6)	0.59 (0.38-0.91)
GEJ	24	12 (50.0)	0.68 (0.21-2.15)
Histological subtype			
Intestinal	127	69 (54.3)	0.65 (0.39-1.07)
Diffuse	46	25 (54.3)	0.38 (0.17-0.86)
No. of metastatic sites			
<2	34	15 (44.1)	0.40 (0.13-1.23)
≥2	153	86 (56.2)	0.61 (0.39-0.95)
Previous total gastrectomy			
Yes	31	15 (48.4)	0.16 (0.05-0.47)
No	156	86 (55.1)	0.77 (0.49-1.20)



ECOG PS, Eastern Cooperative Oncology Group performance status; GEJ, gastroesophageal junction; HER2, human epidermal growth factor receptor 2; HR, hazard ratio; ICR, independent central review; IHC, immunohistochemistry; ISH, in situ hybridization; OS, overall survival; PC, physician's choice; T-DXd, trastuzumab deruxtecan.

The OS subgroup analyses are for patients in the full analysis set (all randomized patients who received ≥1 dose of study treatment).

Table S1. Overall Safety of Study Treatment.

Type of Adverse Event, n (%)	T-DXd (n = 125)	PC Overall (n = 62)
TEAE	125 (100.0)	61 (98.4)
Drug-related TEAE	122 (97.6)	56 (90.3)
TEAE grade ≥3	107 (85.6)	35 (56.5)
Drug-related TEAE grade ≥3	94 (75.2)	27 (43.5)
Serious TEAEs	55 (44.0)	15 (24.2)
Drug-related serious TEAEs	27 (21.6)	5 (8.1)
TEAEs leading to drug discontinuation	19 (15.2)	4 (6.5)
Drug-related TEAEs leading to drug discontinuation	12 (9.6)	3 (4.8)
TEAEs leading to dose reduction	40 (32.0)	21 (33.9)
Drug-related TEAEs leading to dose reduction	38 (30.4)	21 (33.9)
TEAEs leading to dose interruption	78 (62.4)	23 (37.1)
Drug-related TEAEs leading to dose interruption	64 (51.2)	19 (30.6)
TEAEs leading to death	8 (6.4)	2 (3.2)
Drug-related TEAEs leading to death	1 (0.8)	0

PC, physician's choice; T-DXd, trastuzumab deruxtecan; TEAE, treatment-emergent adverse event.

Table S2. Treatment-Emergent Adverse Events in ≥ 10% of Patients Treated With T-DXd.

Preferred Term, n (%)	T-DXd (n = 125)			PC Overall (n = 62)		
	Grade			Grade		
	Any	3	4	Any	3	4
Nausea	79 (63.2)	6 (4.8)	0	29 (46.8)	1 (1.6)	0
Neutrophil count decreased[a]	79 (63.2)	48 (38.4)	16 (12.8)	22 (35.5)	10 (16.1)	5 (8.1)
Decreased appetite	75 (60.0)	21 (16.8)	0	28 (45.2)	8 (12.9)	0
Anemia[b]	72 (57.6)	47 (37.6)	0	19 (30.6)	13 (21.0)	1 (1.6)
Platelet count decreased[c]	49 (39.2)	12 (9.6)	2 (1.6)	4 (6.5)	1 (1.6)	1 (1.6)
White blood cell count decreased[d]	47 (37.6)	26 (20.8)	0	22 (35.5)	5 (8.1)	2 (3.2)
Malaise	43 (34.4)	1 (0.8)	0	10 (16.1)	0	0
Diarrhea	40 (32.0)	3 (2.4)	0	20 (32.3)	1 (1.6)	0
Vomiting	33 (26.4)	0	0	5 (8.1)	0	0
Constipation	30 (24.0)	0	0	14 (22.6)	0	0
Pyrexia	30 (24.0)	0	0	10 (16.1)	0	0
Alopecia	28 (22.4)	0	0	9 (14.5)	0	0
Fatigue	27 (21.6)	9 (7.2)	0	15 (24.2)	2 (3.2)	0
Lymphocyte count decreased[e]	27 (21.6)	8 (6.4)	6 (4.8)	2 (3.2)	0	1 (1.6)
Hypoalbuminemia	18 (14.4)	4 (3.2)	0	8 (12.9)	3 (4.8)	0
Abdominal pain[f]	17 (13.6)	1 (0.8)	0	9 (14.5)	2 (3.2)	0
Stomatitis[g]	14 (11.2)	2 (1.6)	0	3 (4.8)	0	0
Edema peripheral	13 (10.4)	0	0	0	0	0

PC, physician's choice; T-DXd, trastuzumab deruxtecan.

No additional treatment-emergent adverse events were observed in ≥20% of patients receiving PC.

[a] Includes preferred terms "neutrophil count decreased" and "neutropenia."

[b] Includes preferred terms "hematocrit decreased," "hemoglobin decreased," "red blood cell count decreased," and "anemia."

[c] Includes preferred terms "platelet count decreased" and "thrombocytopenia."

[d] Includes preferred terms “white blood cell count decreased” and “leukopenia.”

[e] Includes preferred terms “lymphocyte count decreased” and “lymphopenia.”

[f] Includes preferred terms “abdominal discomfort,” “abdominal pain,” “abdominal pain lower,” and “abdominal pain upper.”

[g] Includes preferred terms “stomatitis,” “aphthous ulcer,” “mouth ulceration,” “oral mucosa erosion,” and “oral mucosal blistering.”